Alex Tsvetanov

Software Engineer

12 Aug 2002

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Alex-Tsvetanov

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Personal Webpage

Programming Languages

- **C++17/20/23**
- **Python**
- **Java**
- **C#.NET**
- **TypeScript**
- **Golang**

Systems & APIs

IOCP • io_uring • kqueue
multithreading
protocols (HTTP & gRPC)
synchronization • SIMD

Tools & Environments

Linux • Windows • macOS
Kubernetes • Docker
CMake • Git • SonarQube
strace • wrk • ab • CI/CD

Domains

Algorithms & Data Structures
Concurrency Design
Distributed Systems
System Architecture
Low-latency Optimization
Native Language Bindings

Soft skills

Strategic Planning
Leadership
Active Listening
Public Speaking
Learning Agility
Resilience
Problem-solving
Critical Thinking
Teamwork
Continuous Learning
Accountability
Initiative

Summary

High-Performance Systems Engineer specializing in C++20 concurrency and cross-platform I/O systems for Windows, Linux, and macOS. Experienced in distributed system design, low-latency optimization, and production deployment. Recognized for debugging complex concurrency issues, collaborating across teams, and continuously improving performance and scalability.

Experience

July 2025 – Present
Senior C++ Engineer
Acronis · Full-time · Hybrid · Sofia, Bulgaria

- Led a team of 4 to design and ship the Disaster Recovery (DR) engine for Cyber Frame and VHI, enabling natively integrated, secure IaaS resilience across 25 global data centers.
- Orchestrated licensing synchronization with Virtuozzo's Key Admin API, aligning complex feature requirements with backend capabilities to ensure predictable, ingress-free infrastructure pricing.
- Engineered high-performance C++ modules for data serialization and integrity, integrating with Golang microservices optimized to handle multi-terabyte Kubernetes clusters per user, meeting core scalability targets for next-generation backup operations.
- Implemented the cyber-pilot SDLC kit to drive AI-adoption within internal projects, streamlining the development lifecycle from business requirement analysis to low-level implementation.

July 2023 – July 2025
C++ Engineer
Chaos Group · Full-time · Hybrid · Sofia, Bulgaria

- Achieved 5x viewport speedup by re-engineering point-cloud transformations using xsimd for cross-platform SIMD abstraction (SSE/AVX-512).
- Architected and open-sourced the "sketchup-cpp-api" library to standardize plugin stability and eliminate cross-process memory corruption.
- Refactored rendering subsystems for thread safety and cache locality, eliminating high-frequency crashes across varied hardware and scene scales.

Sep 2022 – July 2023
Software Architect
SCIA · Full-time · Remote · Leuven, Belgium

- Designed the migration of monolithic Visual C++ logic from legacy COM services to a modern gRPC and Redis-based architecture.
- Modernized the global build system using CMake and vcpkg, unblocking cross-platform development for legacy structural engineering software.
- Directed the evaluation and adoption of gRPC to replace aging internal protocols, ensuring long-term system scalability and interoperability.

July 2022 – Sep 2022
C++ Back-end Developer
PokerStars · Full-time · Hybrid · Sofia, Bulgaria

- Developed real-time limit-checking engines for a global platform peaking at 55,000+ concurrent players.
- Implemented backend logic for localized regulatory compliance and addiction prevention across multiple international jurisdictions.
- Engineered high-availability promotional systems with zero-tolerance for database inconsistency or latency in transaction-heavy environments.

June 2021 – July 2022
Full-stack Developer
VMware · Full-time · Hybrid · Sofia, Bulgaria

- Optimized Solution Builder SQL queries, reducing execution from 30s to 1.5s and resolving systemic deadlocks for thousands of users.
- Delivered Spring Boot/Angular features for enterprise service bundling, managing complex role-pairing and automated quote generation.
- Acted as the primary technical resource for front-end performance and styling, mentoring the team on UI best practices and modular Angular components.

May 2019 – Sep 2019
C++ Software Developer
Codix · Part-time · On-site · Sofia, Bulgaria

- Developed core financial modules including lending, factoring, and debt collection for 9 international Tier-1 banking and financial institutions.
- Maintained and extended high-stakes C++ "night jobs" and background processing services, ensuring data integrity for global financial close cycles.
- Implemented 100+ feature extensions within a shared codebase, balancing client-specific requirements with a unified product architecture.

References

While in high school, Alex's interests focused on AI and Society 5.0. His innovative thinking led us to include him in a specialized young researchers' team for our studies on "Society 5.0 towards 2030". His AI results were further extended for a virtual teacher, providing a visionary outlook for the future of smart education. Alex is a fast-progressing and highly innovative researcher.

Assoc. Prof. Zlatogor Minchev
JTSAC Director
IICT-BAS, March 2026

Alex consistently demonstrates outstanding technical skills and a fearless approach to complex challenges. During our time together, he was instrumental in implementing key features and driving significant infrastructure improvements, directly contributing to the success and stability of our projects.

Noel Warren
Senior Software Developer
Chaos, June 2025

As part of the Architecture team, whose main responsibility was to prototype and introduce innovations, Alex Tsvetanov excelled as a fast learner. No matter which technology we picked (gRPC, Redis, vcpkg), he was always able to quickly learn it and directly build a fully functional prototype. And that was never easy, as he often needed to combine code that was tens of years old with recent technology that was not even suitable for our target platform.

Jan Stříteský
Lead Software Architect
SCIA, April 2023

Alex is not only an excellent programmer, but a sociable and motivated colleague with whom it is pleasant to communicate. He is extremely organized and can be relied upon to learn quickly and clearly, as well as to perform complex tasks independently. I am impressed with his attitude towards the work and the quality results he has achieved.

Irina Drencheva
Software Dev Manager
Codix, October 2019

Education

Sep 2023 – June 2026
Computer and Software Engineering
B.Eng., State-funded · Technical University of Sofia, Bulgaria
Thesis: "Design and Development of a Compile-time ORM library in C++"
Full-time, Grade: Excellent, 240 ECTS.

Sep 2022 – June 2023
Engineering Technology
B.Eng., Self-funded · Katholieke Universiteit Leuven, Belgium
Full-time, Terminated program in English.

Sep 2021 – June 2022
Industrial Engineering in English
B.Eng., State-funded · Technical University of Sofia, Bulgaria
Full-time, Transferred abroad.

Sep 2013 – June 2021
Mathematics & Informatics
High School, State-funded · Sofia High School of Mathematics, Bulgaria
Full-time, Grade: Excellent.

Trainings

June 2016 – Sep 2019
Summer Research School (on-site, Blagoevgrad, Bulgaria)
Summer School · Bulgarian Academy of Sciences
Research-focused program led by top scientists in Europe. Participated in cybersecurity, AI, and systems modeling projects under academic mentorship.

Aug 2018 – Sep 2018
SAP Geeky Camp 5.0 (on-site, Sofia, Bulgaria)
Summer School · SAP Bulgaria
Selective summer tech camp with intensive focus on algorithms, data structures, and solving real-world software engineering problems in teams.

Sep 2017 – June 2018
Java Web Development (on-site, Sofia, Bulgaria)
Telerik School Academy · Telerik Academy
Built full-stack web apps using Java and the Spring Framework. Focused on MVC architecture, dependency injection, and database connectivity.

Sep 2016 – June 2017
C#.NET (on-site, Sofia, Bulgaria)
Telerik Software Academy · Telerik Academy By Progress
Focused on web development with ASP.NET. Covered server-side logic, MVC patterns, and RESTful services with C#.

Sep 2016 – June 2017
Web Development (on-site, Sofia, Bulgaria)
Telerik Software Academy · Telerik Academy By Progress
Developed full-stack web applications using Node.js, HTML, CSS, and vanilla JavaScript. Emphasized frontend-backend interaction and client-server architecture.

Oct 2014 – June 2017
Algorithms & Data Structures (on-site, Sofia, Bulgaria)
Telerik Algo Academy · Telerik Academy
Advanced training in algorithmic techniques, including recursion, dynamic programming, graphs, and greedy algorithms. Participated in national coding competitions.

Sep 2014 – June 2018
C++ & Algorithms (on-site, Sofia, Bulgaria)
Telerik Kids Academy · Telerik Academy
Hands-on programming fundamentals and problem-solving using C++. Built strong foundations in syntax, logic, and algorithmic thinking from an early age.

Publications

Nov 2025
Optimizing Asynchronous Event Dispatch in Modern C++ Publish/Subscribe Systems
TELECOM 2025, Sofia, Bulgaria • 10.1109/TELECOM66943.2025.11304088
Alex Tsvetanov

June 2025
Modern C++ Publish/Subscribe Pattern: Design, Challenges, and Implementation
ICEST 2025, Ohrid, North Macedonia • 10.1109/icest66328.2025.11098436
Alex Tsvetanov • Assoc. Prof. Ivan Stankov, Ph.D.

Nov 2024
Matching Text from Start to Finish Against Multiple Regular Expressions
TELECOM 2024, Sofia, Bulgaria • 10.1109/telecom63374.2024.10812214
Alex Tsvetanov • Assoc. Prof. Ivan Stankov, Ph.D.

Nov 2017
An Analytical Outlook towards Digital Training Security Transformations
ICAICTSEE-2017, Sofia, Bulgaria • 10.13140/RG.2.2.20333.28645
Assoc. Prof. Zlatogor Minchev • Luben Boyanov, PhD • Antoan Georgiev • Alex Tsvetanov

Certificates & Grants

Oct 2025 1st place among the university teams in the cybersecurity hackathon "Big SCHWARZ Interlude"
Sofia, Buglaria

Oct 2019 Runner-up in President's awards "John Atanasoff"
Sofia, Buglaria

Sep 2019 Top 10 with Headstarter, "Best Youth Startup 2019" by the Youth Entrepreneurship Accelerator Ustvarjalnik
Ljubljana, Slovenia

May 2019 1st place with Headstarter, National Entrepreneurship Competition by Teenovator
Sofia, Buglaria

Mar 2019 Gold medal in Sofia Open Algorithmic Programming Competition
Sofia, Buglaria

Mar 2018 Bronze medal in Softuniada Competition in Algorithmic Programming
Sofia, Buglaria

Nov 2017 Bronze medal in International Competition in Algorithmic Programming "John Atanasoff"
Sofia, Buglaria

Jan 2017 Silver medals in U16 of Softuniada Competition in both Algorithmic Programming and Web and Software Projects
Sofia, Buglaria

Languages

BG Native ●●●●●

EN C1 (SAT) ●●●●○

Hobbies

Podcasts about A(u)DHD
Brain Health • Gym
Football • Table tennis
Rock&Metal Music

My mindset

If it breaks under pressure, it was already broken.

Find me

alex.tsvetanov.2002

tsvetanov__alex

Projects

2017 – Present — **Coroute/WebFrame++** (part of a patent application)
Cross-Platform Web Framework · Personal
- Designed and benchmarked a high-performance asynchronous server architecture handling 750'000 to 1.2 million requests per second across Windows and Linux (measured with wrk and ab).
- Implemented a coroutine-based I/O scheduler and lock-free queues, improving throughput by about 13%.
- Developed a modular networking layer that abstracts IOCP, io_uring, and kqueue into a unified asynchronous API.
- Evaluated coroutine scheduling models and caching strategies to minimize latency jitter, and extended the framework with HTTP pipelining and asynchronous file streaming.

2018 – 2022 — **Headstarter**
High-school Entrepreneurship Project · Team
- Developing and maintaining the Headstarter web platform for connecting young students with businesses offering internships and jobs.
- Supporting the platform goals of guiding students in career paths, networking, and gaining experience.
- Organizing the team's workload for efficient operations and service delivery.

2016 – 2019 — **TechEdu++**
Learning Management System for Informatics · Team
- Developing a learning management system (LMS) with Informatics courses for high school students.
- Creating short, engaging video lessons in multiple programming languages.
- Integrating TechEdu++ into the teaching processes at Sofia High School of Mathematics.

Volunteering

Nov 2025 — **Lighting talk speaker at OpenFest** (on-site, Sofia, Bulgaria)
Delivered a 5-minute lightning talk on leveraging C++20 coroutines (`co_await`) over unified kernel I/O queues, covering the submit/completion queue model, a coroutine-friendly async I/O API, and a thread-pool executor design — demonstrating how modern C++ enables high-throughput, low-latency I/O with synchronous-style code.

Apr 2018 – Apr 2020 — **Author & Co-organizer of Open Doors Day at Sofia High School of Mathematics** (on-site, Sofia, Bulgaria)
Organized 3 annual editions of the Open Doors Day event, where 19 alumni shared their career journeys post-graduation. Coordinated all logistics and communication with guest speakers, created the event schedule, and set up the server infrastructure for live streaming and online attendance.

Sep 2017 – June 2018 — **Student Mentor at Dojo@TelerikAcademy** (on-site, Sofia, Bulgaria)
Volunteered as a mentor guiding young learners ("Ninjas") through basic programming concepts. Assisted them in building small personal projects, encouraged curiosity, and provided support during hands-on coding sessions.

May 2017 – June 2017 — **Technical Trainer at Software University** (on-site, Sofia, Bulgaria)
Delivered Python programming lectures to a diverse group of learners, including working professionals retraining for software roles. Led live coding sessions, assigned practical exercises and homework, and evaluated performance through final exams. Adapted teaching style to accommodate varying experience levels and backgrounds.

Sep 2016 – June 2017 — **Project "Testrain" part of the "A Computer Simulation Virtual Test of Silvan Tomkins' Affect and Script Theory" conference paper** (on-site, Sofia, Bulgaria)
Developed the web interface for an experimental platform simulating emotional response theory based on Silvan Tomkins' work. Contributed to the visualization and interaction layer of the research tool used in academic experiments.

Sep 2013 – June 2021 — **Competitor from the Sofia city team in Algorithmic programming** (on-site, Sofia, Bulgaria)
Competed in national and international algorithmic competitions over 8 years, solving challenging problems within limited amount of time using advanced algorithms, data structures, and analytical thinking. Won numerous awards at both levels, becoming the youngest competitor on record at age 9 and the youngest gold medalist at age 10.